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Integration of Knowledge Based Approach and Multi-Criteria Optimization in Multi-Disciplinary Machine Design

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Abstract. The approach presented in the paper attempts to integrate the concept of a knowledge based software (knowledge storage/management software) into a multi-criteria optimization applied for a multi-disciplinary problem. The approach focuses on the management issues of a multi-stage solving procedure and its functionalities.

Keywords. Engineering knowledge modelling, multi-criteria optimization, multi-disciplinary design

Introduction

The approach presented in the paper attempts to integrate the concept of a knowledge based software (knowledge storage/management software) into a multi-criteria optimization applied for a multi-disciplinary problem [1][2][3][4][5][6]. The special focus is put on the management issues of a multi-stage solving procedure and its functionalities [3].

The area of multi-disciplinary optimization applications is slowly but continuously extending [1][7]. It is no longer limited to astronautics and aviation. This comment doesn't only concern a research but also an industry.

One unexpected phenomenon which appeared while applying multi-disciplinary optimization in industry was the negative attitude of industrial engineers [1][3][7]. They had objections against the high costs and risks connected with those tools. The high cost and high risk result from the fact that introducing multi-disciplinary optimization to a new domain of application needs developing sets of new software solutions which are necessary to support the engineering activities.

At the initial stage of optimization process it is often difficult to specify and foresee the range and size of the environments which have to be modeled. The labor – intensive analyses become necessary and conclusions have to be formulated concerning the systems, their modules and the tools which shall be integrated. Each made decision is associated with concrete cost and risk.

Multi-disciplinary models are relatively big and are very complex [8][9][10][11][12][13]. They are assemblies of partial models which have their origin in different disciplines. Linkages of partial models reflect the fact that products and problems of their designing are complexes which are homogenous and multi-aspect.

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The merit knowledge of the designers decides which models shall be applied in the certain case [7][14][15]. Usually, the members of a design team are quite familiar the professional capabilities of their colleagues from the same design office or design team [10] [14]. They also know the possibilities of the used tools and methods. Nowadays those capabilities and possibilities play an important role in the design process, even in small firms, because the realised project must meet the requirements of the market. Such knowledge should be acquired, stored and re-used.

In the next chapters multi-disciplinary design problems, aspects of the design knowledge associated with them, concepts of tools for their knowledge management are characterized. It is assumed that the multi-disciplinary optimization approach will be used by a small design office.

1. Characteristics of multi-disciplinary problem modeling

Designers solving machine design problems try to build computer models [13][16] [17][18][19], examine, verify and validate them (Figure 1). Each of the activities in those models has its knowledge sources' background [2][13][16][17]. In most cases this process doesn't end up with one sequence of well selected activities. Designers' work also consists of a series of corrections where each correction gives a new variant (or variants) of the considered model together with its examination and verification [2][3]. Between two corrections (due one last- examined and the one just being created) the designer performs some intellectual work, considering what to do next. Designers perceive the whole path made of separate corrections, together with its mental interruptions, as one reasoning process, which finally leads to one satisfying solution. This process represents a kind of real-life optimization. If this process is handled in a computer environment we speak of knowledge based approach.

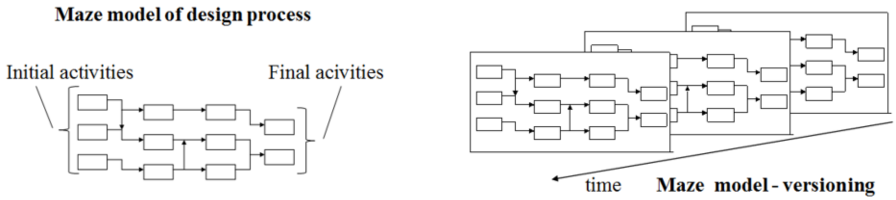

Figure 1. Personal activities and their structuring

On the other hand we shouldn't forget that decision problems solved by designers, in most cases, look like a chain of interconnected single decision problems with loops, iterations, macro-decisions etc. [3][4][5] (Figure 2).

Formal multi-criteria optimization methods (especially based on multi-criteria, multi-level hierarchic approach [2][3][20][22][23]) can support the designer (or the designers in case of collaboration) in his decision making process [20][22][23][24] (Figure 3). However, the multi-stage, knowledge based nature of the decision making process will remain the same.

The designer exploits the results of a certain examined decision problem at the further stages of the design process [2][4][16]. The interactions between the different stages of the design process can be informal or formal (Figure 4). In the first case it is difficult to build a formal integrated approach. In the second case, however, we can consider interactions between different multi-criteria problems solved by the designer on the realized design process path.

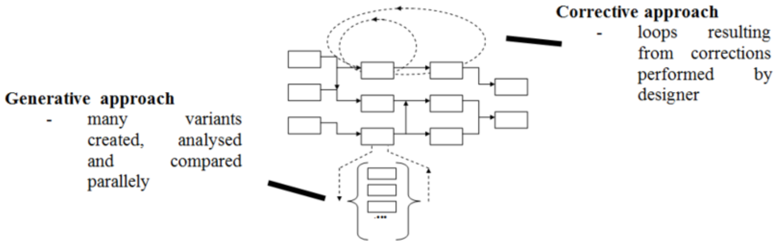

Figure 2. Corrective and generative way of functioning of performed processes.

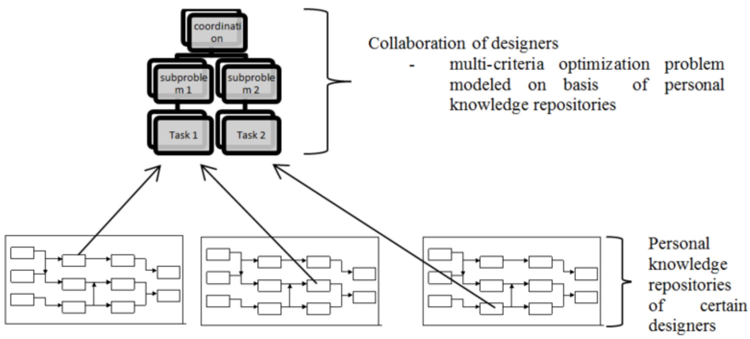

Figure 3. Multi-disciplinary problem modeling on the basis of personal knowledge repositories.

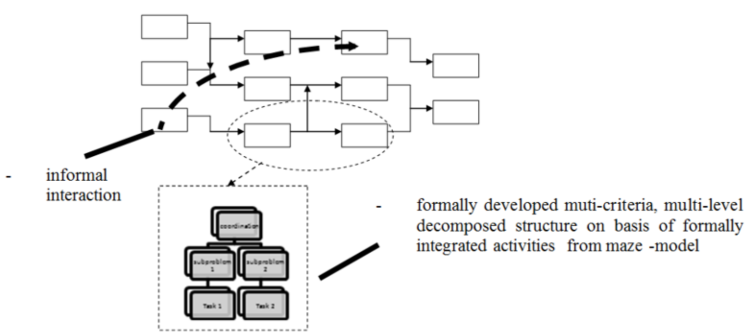

Figure 4. Informal and formal connections between activities.

Our chances of finding new and better results increase when we examine problems which have never been optimized (even without optimization methods) [7][15][20][22]. Such a task is typical for a multi-discipline design approach. Looking at the history of such an approach it seems that the different disciplines evolved separately. But in reality, in the case of real technical objects (from the so called product perspective), disciplines are integrated and overlap [8][9][25].

For a human the abilities to specialize in several disciplines are very limited. Therefore with a complex engineering problem a group of specialists has to cooperate as one team. If each team member optimizes his problem as a sequence of model corrections in a knowledge based approach or uses optimization methods and if all the team members cooperate, then the problem arises who dictates the parameters and makes the necessary decisions [1][4][5][7][9][14][16].

In real-life situations there is always a position of a leader who makes the decisions and shares the resources.

In the case of a multi-person team this means (collaborative design) the leader of the whole complex problem has a plurality of designers and he can therefore compose a path made of problems which are solved by different designers [1][7][10][19].

The development of multi-disciplinary models can also include considerations about mutual linkages (between partial problems) which have been never noticed before. Those linkages may exist between disciplines [14][26] or they can go deeper – noticing situations where disciplines overlap and mentioned problems are not separable [8][9].

2. Computer support of engineering design - example

The chapter presents an example of the dedicated computer environment to support the modeling of multi-criteria optimization problems together with the storage and management of their corresponding engineering knowledge as well as their corresponding data management.

The example deals with the problem of spiral stairs design [27]. External requirements decide about the general structure of the stairs. Because of that only limited changes of the structure are possible (Figure 5).

When setting about a multicriteria optimization, the respective problem (the stair in our case) is formulated with the goal to achieve its most optimal structure and the most optimal values of parameters.

This is the classical way a conceptual design problem is solved. When a conceptual solution has been found, detailed solutions for the whole project can be generated. The second stage of the prosess is deeply integrated with the optimisation models. With the final results the manufacturing process can be provided. The second module cooperates directly with a CAD system.

Similar ways of solving engineering problems can also be found in other areas. For instance: defining the structure of the car [20][22][28][29](Figure 5). For many automotive units exists only limited numbers of variants. In those cases, approaches based on Case Based Reasoning dominate [2]. There are also tasks with numerous variants – for instance: the configuration of the seats layout in a car or bus, or the routing of pipes and cables, etc. For such problems, the optimization approach is the most suitable. The example of the bus is presented further in the section 5.

Stairs structure

- structural stairs variant – many solutions
- barriers – a few variants
- mounting connections – a few variants

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Bus structure

- engine chamber – a few variants
- driver cockpit – a few variants
- passenger seats distribution – many solutions

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Figure 5. Structure of variants of certain products.

3. Integration of knowledge based approach with multi-criteria optimization in engineering design

Engineering design is a multi-stage process built interactively by designers [2][16]. Design processes realized in industry are characterized by a relatively little repeatability which results from a continuous engineering knowledge development, changing external requirements and a growing complexity practically solved tasks.

The knowledge owned and used by a designer can help to make certain inferences which may yield new elements in the realized design process of a certain product. When working the active designers look for the suitable knowledge, and often acquire that knowledge from many sources [2][16][17]. This is a kind of optimization with predefined goal areas. The fixed goals are achieved by searching for already existing similar solutions and their adaptations (Case Based Reasoning) [2][3] (Figure 6).

Process of building of MCDSS application

- Designer and his professional knowledge
- Selection of multi-objective or multi-attribute approach
- Building concept of modeling risk and uncertainty
- Selection of decision variables, constraints and criteria
- Identification of decision maker's (designer) preference model
- Selection of suitable multi-criteria decision making methods
- Building concept of integrating CAD/CAE system with MCDSS application

Process of building of CBR application

- Designer and his professional knowledge
- Building concept of case
- Building concept of similarity measure
- Building concept of integrating CAD/CAE system with CBR application
- Building concept of case adaptation module

Figure 6. Core processes of building Multi-Criteria Decision Support System/ Case Based Reasoning (MCDSS/CBR) applications.

With many design problems the inferencing actions lead to design problems with numerous number of variants, modeled formally explicitly or implicitly. The selection of the most preferred solution with the help of inferencing approaches often costs a lot of work. In such a situation multi-criteria optimization brings more fruitful results (Figure 6).

Models developed formally as partial problems belonging to different disciplines can be analyzed by many different computer tools. The analysis of the performed partial problem can be done separately or can have common elements: decision variables, state variables, other (coordination) variables, etc. The integrated sets of partial problems can be solved as one large multi-disciplinary optimization problem. As a final result a solution is obtained which may be optimal from a global perspective.

4. Concept of engineering design knowledge modeling

Design processes are permanently modified and improved [10][16][18][2]. Those processes are performed on the basis of the designers' knowledge and cause a growing complexity of the respectve design tasks [24][26] as the applied knowledge concerns large fields of reality. In many cases the conducted design processes are unique.

One of the biggest challenges is the dilemma how to model multi-disciplinary design issues with limited human, hardware, software and financial resources (assumption concerning small company circumstances [3][7][15]). Like always the most optimal result is obtained after many iterations, lots of experiments and extensive consulting and knowledge sharing. Because of that effort any progress achieved that way is worth storing and re-using in the future. The knowledge which stays behind that progress should be stored in special repositories (Figure 7).

A data base is a helpful tool during the process of multi-disciplinary optimization in engineering design modeling [3][30][31][32]. The data base can store the following resources:

1) General description of the multi-disciplinary problem; problem description, list of partial problems, structure, theoretical and formal background; personal data of modeling persons,
2) Detailed description of the partial problems; description of variables, description of resulting variables, the description of methods and tools; the data concerning versions of the applied software,
3) For the partial problems; information concerning test iterations, information concerning model improvements and its decision background,
4) For the multi-disciplinary problem; information about integrated partial problems, information about performed iterations and their results,

The created software can store extensive, complete information about the evolution of the formal problem modeling and the evolution of the formal and informal structured problem's knowledge background - together with its real-life facts and observations, together with its associated hypothesises, mental or mental-formal, partial or complete multi-disciplinary models.

5. Example

In the chapter the example of the computer system supporting the process of the bus structure definition is presented. The example is based on the cooperation with a known bus manufacturer.

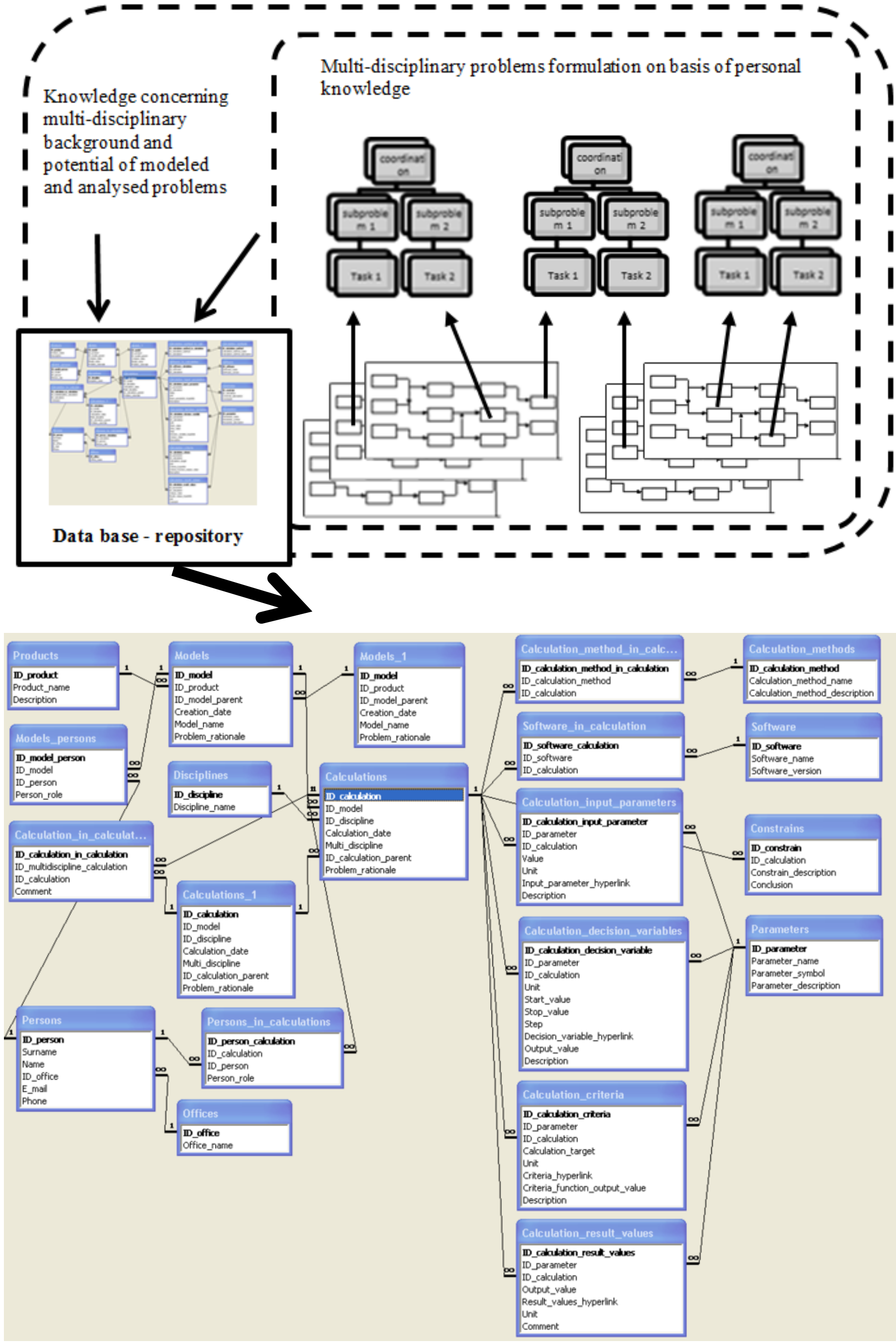

Figure 7. Concept of repository of knowledge concerning formal componets of multisciplinary optimization and its personal formal and informal knowledge background.

It shows modules for the configuration of the whole bus structure (Figure 8) together with its seats layout and attempts of seats layout optimization (Figure 9). The developed system is equipped with tools for the fast presentation of alternative variants.

For many bus modules exist only limited numbers of variants (for instance for the cockpit, the engine chamber, etc.). For those cases, an approach based on Case Based Reasoning is proposed in the system (Figure 10).

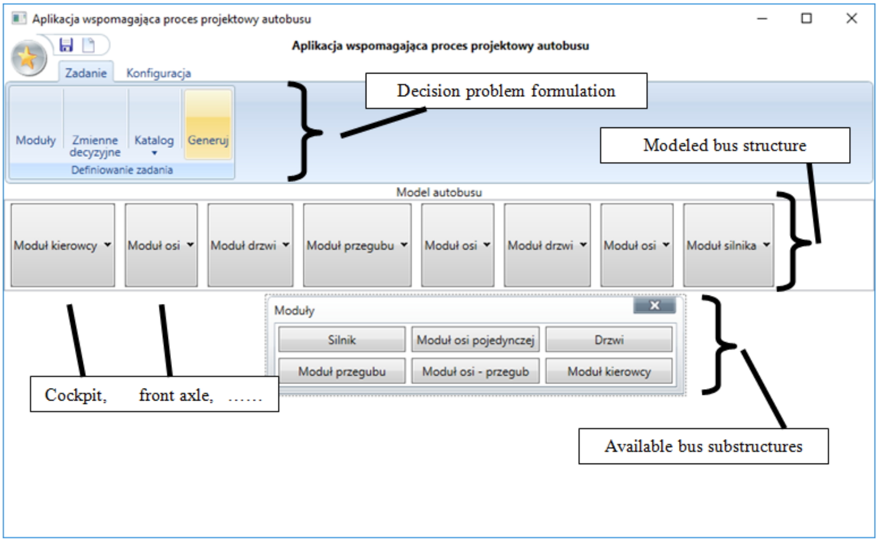

Figure 8. Tools for bus structure definition and decision problem formulation.

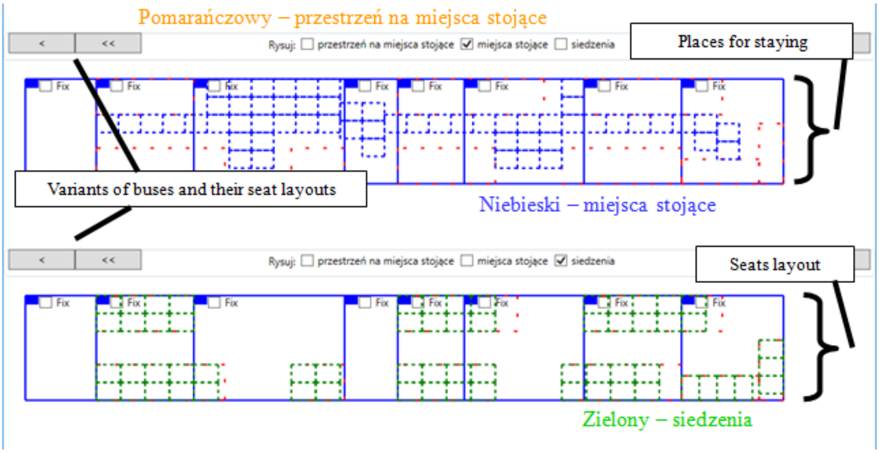

Figure 9. Results of automatic bus structure and seats layout generation.

There are also tasks based on Knowledge Based Engineering (KBE) where the developed system allows to create a sets of data for the prepared dedicated, KBE, problem-oriented 3D models.

The functionalities offered by the developed system can be used in sequences and give some knowledge and practical experience. The results of these processes can be modeled and stored in the environment – the dedicated database is shown in Figure 7.

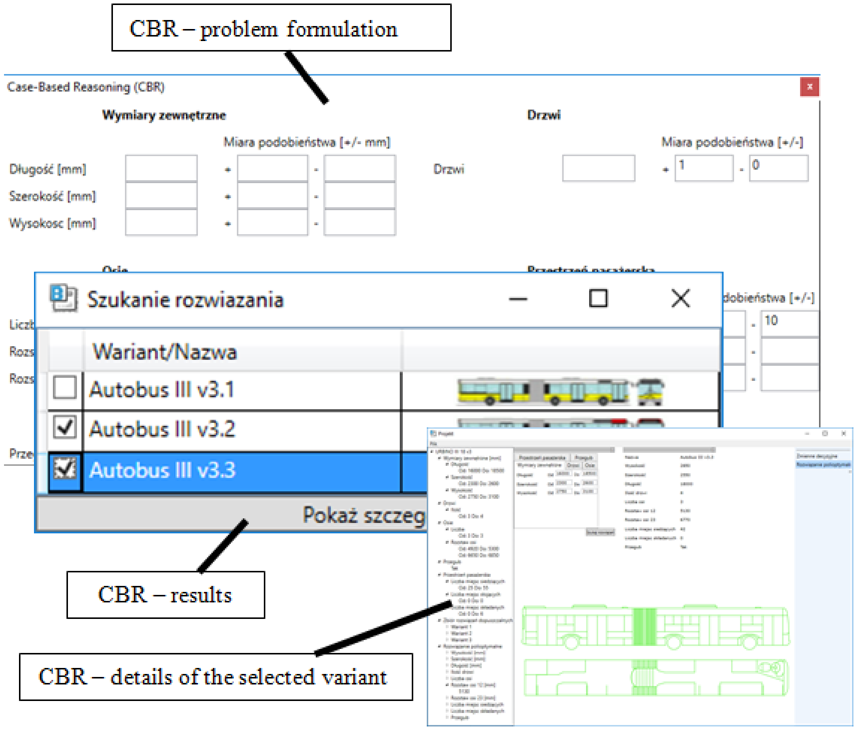

Figure 10. Case Based Reasoning functionality applied in case of the whole bus.

6. Conclusions

The paper presents the concept of a software for supporting the process of engineering knowledge storing in case of multi-disciplinary design tasks modeling and development.

The software is created to store information about the evolution of the problem modeling and the associated knowledge resources.

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